Arista Access Node BTM-ACC-19 Optical Transport Failure SOP

Node: BTM-ACC-19 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_8549

Description

This SOP addresses a warning-level Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Arista Access Node BTM-ACC-19. The failure is attributed to the Optical Transmission Media (OTM) degradation or cuts in the optical fiber. The OTM is a critical component in the Arista Access Node's optical transport chain, facilitating high-speed communication between nodes through the use of SFP/XFP transceivers. The SFP/XFP transceivers are responsible for transmitting and receiving optical signals over the fiber, and any degradation or loss of signal due to cuts or OTM issues may impact network performance. Furthermore, the SFP/XFP transceivers are connected to the Arista Access Node's Line Card, which processes the optical signals and converts them into electrical signals for network processing. In the event of a failure, the Line Card may not function correctly, resulting in a warning-level Optical Transport Failure.

Pre-requisite Checks

- Check: Verify the site location BTM Layout Aggregation is accessible by remote hands or on-site personnel.

- Check: Verify the redundant link status of the Arista Access Node BTM-ACC-19 to ensure network redundancy is maintained in case of a failure.

- Check: Verify the environmental conditions at the site location, including temperature, humidity, and cleanliness, to ensure optimal operation of the Arista Access Node.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation or failure	High
Optical fiber cuts or damage	Medium
Optical transmission media (OTM) degradation or damage	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces status, Expected Output: Interface Status: 1/1/1 (SFP/XFP) is down, Description: This command will verify the status of the SFP/XFP interface and identify if it is down due to the Optical Transport Failure.

Command: show interface 1/1/1 detail, Expected Output: SFP/XFP is down due to optical signal loss, Description: This command will provide detailed information about the SFP/XFP interface, including the reason for the failure, which in this case is optical signal loss.
```

```
Command: show optics 1/1/1, Expected Output: SFP/XFP is degraded due to high optical power levels, Description: This command will provide information about the SFP/XFP's optical characteristics, including power levels, and identify if it is degraded due to high power levels.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver, Command: no shutdown; no module 1/1/1; install-transceiver 1/1/1 new-sfp, Description: Replace the SFP/XFP transceiver with a new one to resolve the Optical Transport Failure. This step requires a physical replacement of the transceiver, which may require remote hands or on-site personnel.

Step 2: Step: Verify the optical fiber connections, Command: show optics 1/1/1 detail, Description: Verify the optical fiber connections and ensure they are secure and not damaged. This step requires a visual inspection of the fiber connections.

Step 3: Step: Reset the Line Card, Command: hw-module module 1/1/1 reset, Description: Reset the Line Card to resolve any configuration issues or software problems that may be contributing to the Optical Transport Failure.

Post-Check Verification

```
Command: show interfaces status, Expected Output: Interface Status: 1/1/1 (SFP/XFP) is up, Description: Verify that the SFP/XFP interface is up and functioning correctly after the resolution steps.

Command: show optics 1/1/1 detail, Expected Output: SFP/XFP is functioning correctly with normal optical power levels, Description: Verify that the SFP/XFP is functioning correctly with normal optical power levels and without any degradation or loss of signal.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: copy running-config startup-config, Description: Restore the original configuration to resolve any issues caused by the resolution steps.

- Step: Reboot the Arista Access Node, Command: reload, Description: Reboot the Arista Access Node to resolve any issues caused by the resolution steps.